ABDULAZIZ AL OTAIBI

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EDUCATION

The Pennsylvania State University, University Park, PA **December 2026**
Bachelor of Science, Mechanical Engineering *Cumulative GPA: 3.65 / 4.00*
Major GPA: 3.92 / 4.00

Relevant Coursework: Introductory Microeconomic Analysis and Policy, Design Methodology, Product Design and Manufacturing Processes, Mechatronics, Vibration of Mechanical Systems, Thermodynamics, Fluid Mechanics, Heat Transfer

PROJECTS

e-Formula Car Body Aerodynamic Optimization **Spring 2026**
Penn State Design Competition, ME 340 Design Methodology

- Advanced to the quarterfinals of a semester long competition against 64 teams, judged on measured lap performance.
- Developed three original body designs beyond the provided baseline in SolidWorks, iterating geometry to reduce aerodynamic drag and increase downforce at sustained high speed.
- Validated each iteration through CFD analysis and wind tunnel testing at 201 mph simulated airflow, then printed and dimensionally verified every build to a 155.1 mm final body specification.
- Timed the final design over a 20 m three lap course for a best run of 4.55 s, reconciling simulated and experimental results to select the winning concept.

Pendulum Tuned Mass Damper Design **Summer 2026**
ME 370 Vibration of Mechanical Systems

- Modeled a four degree of freedom building and damper system in MATLAB and ran a parametric grid search across the full design space to identify the configuration meeting a 7 cm peak displacement constraint under impulse loading.
- Reduced a nonlinear dynamic system to a tractable linear model, making an exhaustive sweep feasible within the project timeline rather than testing a handful of candidate designs.

Autonomous Robotics Control System **Spring 2026**
ME 454 Mechatronics

- Built and tuned closed loop PD, PID, and finite state machine controllers in C++ for wall tracking, precision parking, and emergency braking on an embedded robotics platform.
- Isolated a motor wiring asymmetry through systematic elimination testing after repeated failed trials, restoring consistent performance across multiple hardware units under deadline.

Integrated Speaker and Desk Lamp Assembly **Fall 2022**
EDSGN 100 Cornerstone Design

- Modeled the speaker body, perforated grille, controls, and gooseneck lamp arm as separate SolidWorks parts and mated them into one assembly, balancing two competing functions in a single desk sized form.

LEADERSHIP AND ACTIVITIES

Competitive Table Tennis, The Pennsylvania State University **Fall 2022 to Present**

- Ranked among the top 10 table tennis players at Penn State University Park through sustained tournament competition.

Public Speaking, CAS 100A Effective Speech **Spring 2023**
The Pennsylvania State University

- Delivered weekly speeches to a live audience of 25, building structured delivery and audience engagement.
- Completed an impromptu assignment allowing one minute of preparation, organizing and presenting ideas under time pressure.

SKILLS AND ADDITIONAL

Analytical: Data analysis, parametric modeling, design of experiments, experimental validation, root cause analysis
Software: Microsoft Excel, PowerPoint, MATLAB and Simulink, SolidWorks, C and C++, Arduino IDE
Languages: Arabic (native), English (fluent)